

Module 45 - String Methods

1. What are String Methods?

String methods are built-in JavaScript functions used to perform different operations on strings, such as searching, extracting, modifying, or converting string values.

2. `charAt()` Method

```
let str = "Hello World";  
  
// Returns the character at the specified index.  
  
console.log(str.charAt(str.length - 1));
```

3. `charCodeAt()` Method

```
let str1 = "Javascript";  
  
// Returns the Unicode value of the character at the specified index.  
  
console.log(str1.charCodeAt(6));
```

4. `concat()` Method

```
// Combines two or more strings and returns a new string.  
  
let newStr = str.concat(" ", str1);  
  
console.log(newStr);
```

5. `includes()` Method

```
let str2 = "Single Threaded";  
  
// Checks whether the specified text exists in the string.  
  
console.log(str2.includes("j"));
```

6. `indexOf()` Method

```
let newStr1 = "Kesavan";  
  
// Returns the index of the first occurrence of the specified character.  
  
console.log(newStr1.indexOf("a", 4));
```

7. `lastIndexOf()` Method

```
let newStr2 = "Kesavan";  
  
// Returns the index of the last occurrence of the specified character.  
  
console.log(newStr2.lastIndexOf("a", 4));
```

8. repeat() Method

```
let newStr3 = "Javascript";  
  
// Repeats the string the specified number of times.  
  
console.log(newStr3.repeat(3));
```

9. replace() Method / replaceAll() Method

```
// replace() Method / replaceAll() Method  
let str3 = "Js is a Script language - Js";  
  
// Replaces the first matching occurrence.  
console.log(str3.replace("Js", "Javascript"));  
  
// Replaces all matching occurrences.  
console.log(str3.replaceAll("Js", "Javascript"));
```

10. slice() Method

```
let newStr4 = "Single Thread";  
  
// Extracts a portion of the string from the specified index.  
console.log(newStr4.slice(3));  
  
// Extracts characters from index 3 up to, but not including, index 8.  
console.log(newStr4.slice(3, 8));  
  
// Extracts a portion using negative indexes from the end.  
console.log(newStr4.slice(-3, -1));
```

11. substring() Method / substr() Method

```
let newStr5 = "Single Thread";  
  
// Extracts characters from the specified index to the end.  
console.log(newStr5.substring(3));  
  
// Extracts characters from index 3 up to, but not including, index 8.  
console.log(newStr5.substring(3, 8));  
  
// Negative values are treated as 0 by substring().  
console.log(newStr5.substring(-3));  
  
// If start is greater than end, substring() swaps the two values.  
console.log(newStr5.substring(8, 0));
```

12. split() Method

```
let words = "My Name is Iron-Man";  
// Splits the string into an array using space as the separator.  
console.log(words.split(" "));  
// Splits using space and limits the result to 3 elements.  
console.log(words.split(" ", 3));  
// Splits the string using "-" as the separator.  
console.log(words.split("-"));  
// Splits using "-" and limits the result to 1 element.  
console.log(words.split("-", 1));
```

13. startsWith() Method

```
let words1 = "My Name is Iron-Man";  
// Checks whether the string starts with the specified value.  
console.log(words1.startsWith("M"));  
// Checks whether "N" starts at index 3.  
console.log(words1.startsWith("N", 3));  
// Checks whether "N" starts at index 4.  
console.log(words1.startsWith("N", 4));  
// Checks whether "Name" starts at index 3.  
console.log(words1.startsWith("Name", 3));
```

14. endsWith() Method

```
let words2 = "My Name is Iron-Man";  
// Checks whether the string ends with the specified value.  
console.log(words2.endsWith("n"));  
// Checks whether the string ends with "n" at the specified length.  
console.log(words2.endsWith("n", 5));  
// Checks whether the string ends with "n" using the full string length.  
console.log(words2.endsWith("n", words2.length));  
// Checks whether the first 14 characters end with "Iron".  
console.log(words2.endsWith("Iron", words2.length - 4));  
// Checks whether the first 14 characters end with "Iron".  
console.log(words2.endsWith("Iron", words2.length - 4));
```

15. toLowerCase() Method

```
let words3 = "My Name is Iron-Man";  
// Converts all characters to lowercase.  
console.log(words3.toLowerCase());
```

16. toUpperCase() Method

```
let words4 = "My Name is Iron-Man";  
// Converts all characters to uppercase.  
console.log(words4.toUpperCase());
```

17. trim() Method

```
let words5 = " My Name is Iron-Man ";  
// Removes whitespace from both the beginning and end.  
console.log(words5.trim());
```

18. trimStart() Method / trimLeft() Method

```
let words6 = " My Name is Iron-Man ";  
// Removes whitespace from the beginning.  
console.log(words6.trimStart());  
console.log(words6.trimLeft());
```

19. trimEnd() Method / trimRight() Method

```
let words7 = " My Name is Iron-Man ";  
// Removes whitespace from the end.  
console.log(words7.trimEnd());  
console.log(words7.trimRight());
```